

# A Packing of 5 Four-Spaces in $\mathbb{R}^{12}$

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## Abstract

We give 5 4-dimensional subspaces of  $\mathbb{R}^{12}$  whose smallest squared chordal distance is at least 3.333129. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.199812, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 5 subspaces of dimension 4 in  $\mathbb{R}^{12}$  whose pairwise squared chordal distances  $D(a, b) = 4 - \text{tr}(P_a P_b)$  are all at least*

$$D_{\min} = 3.333129 \dots$$

The exact value of  $D_{\min}$ , a rational number, is given in Appendix A.

## 2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some  $N$  in the row  $m = 12$ ,  $n = 4$ . The cell  $N = 5$  is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.199812, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is  $4(12 - 4)/12 \cdot 5/4 = 3.333333$ , and the value above lies below it.

## 3 Verification

Each subspace is given by an integer  $m \times n$  basis  $Y$  (coordinates times  $10^{12}$ , rounded). With  $G = Y^\top Y$  the projection is  $P = Y G^{-1} Y^\top$ , so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N5.txt`

in <https://github.com/Ge-limin/math719>.

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\*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

## 4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 60 seconds on one laptop, with about ten such searches running at once.

### A Exact value

numerator:

```
333312986601294099796470285368552136251196456279106966237718768800215252
594012014409391084529006871560510189063478820916233318366976578983499639
8911515893433717401899605473788928552041170647699
```

denominator:

```
99999999999844236268532335789620250246259222183149241535384562513866017
181307897506456159229745262042435975770078376611951646863545472567065134
539921615828464411665668789835998995616422047772
```

### B Coordinates

5 blocks of 12 rows with 4 integers each; divide by  $10^{12}$ . File: `grassmann/packings/m12n4N5.txt`.

```
-528627719317 -365314729806 -238223243537 -13049603585281985916 -158292179834 -281942476612 -15793771911493583487 -300179968485 -135458456751 202234948559
-30780768398 -1152314745 115984091846 453917115774 369032705989 216836303280 9542891313 65321844895 -142723247708 -171540542448 650039755627 216405546486
-180366373341 562721229555 -46767534438 54987531125108389781264 269135830065 407583102687 631550448755-13251994532 382209616575 -57115120843 -327380689921
6886026350 -280289957262 92679853045 376902861259 -251790863371 636077907772 -92789691783 -4731161980238185843465 -352309133966 -167374487438 -298011148286
-248489764161 -38752836045 569966762846 -16958350297811472273441 300075937856 -107018192920 2199463292384240466178 358050749112 274519877266 71570749691
349388594234 316971245660 133820872933 -16969861863245059372160 188051385213 105766322749 -63432381175171412048344 -437894038086 -42249298777 15572675470
-442162005529 51329186988 132853257308 -570509391832545124104 1547288341 -17561900574 184534136887 -72866155115 -15307733306 219994808792 -737898676692
-70045340585 17824891543 4924872856 -137600883388 -259730502522 -141566758774 -82109108880 3384082991289497132125 27404218494 326292337993 -107826806729
-40972938578 -403598489705 357483087871 18845417614274490271983 645787541350 24138019871 23830416598 8640913364 110879579761 -47091974702 35053076828
476689518116 -441870515053 -52065671008 27314383812219857647669 41978293366 63250979655 399438647606 89703884027 -42680447982 199504632860 -131580442275
-149465043347 55812659112 386775567926 287078580907-129069443526 211577851028 630886556929 -72955408822648716724 466043907313 11526143843 296543830019
-227026900543 -71727728731 -527708165516 2454175402052927809562 -213519207469 46351168851 -5409915671027374075390 -10497143504 -235667876247 -350586073898
261901093070 -49390483300 -196355602993 28862658233115203996736 -550908108917 117473437082 8652421595291343799143 -237320588194 581748223517 102583582457
-99240394009 -104623947258 -516109224054 1824472027103961363839 -83879665009 637225715123 213138110535-211235701756 -63697617865 114848069839 -668588957393
289180130900 -512710210505 69249716959 -275002866025116983699 -68881487219 -290214073556 43194725085 -237309951946 4795450149 -7504919848 -481086103727
-227420564392 -222122530860 518074273816 -250869143874489502204 173516570140 -247951259973 -34136445788114239084909 760517036611 227882782261 -203241038463
-187429420744 -320328308032 82810686575 -904203634226203151580 -345007163350 139105466350 97970117633-369611996024 -171840623899 397146509882 103333392050
-339042853440 32746319555 139521938371 139657503769-642182071959 301926124853 175500831645 1208501154084082489645 277358523398 374022537313 13950139401
393048222113 75462811270 355558037302 -3894452170794331827176 -98222914647 460317807193 59391126201 160366313963 -120186376058 145765456036 99968788087
-517854233761 -126188163055 121726049609 2434859601783871498061 263601968740 -183763538928 356461423490749321891 -94274395529 424068755762 -139490256057
```